

Geometry

① Point •

② line segment

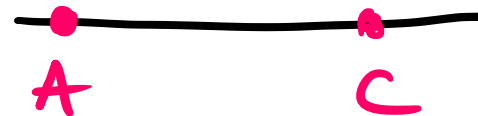


③ Line

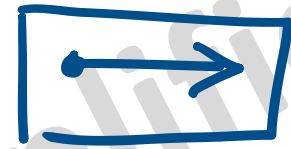


collinear

• B

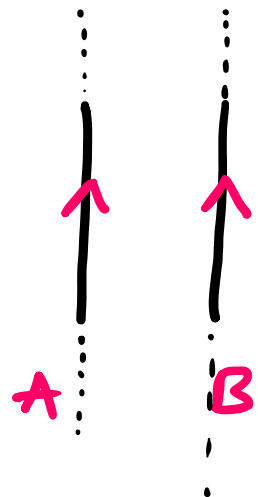


Non-collinear



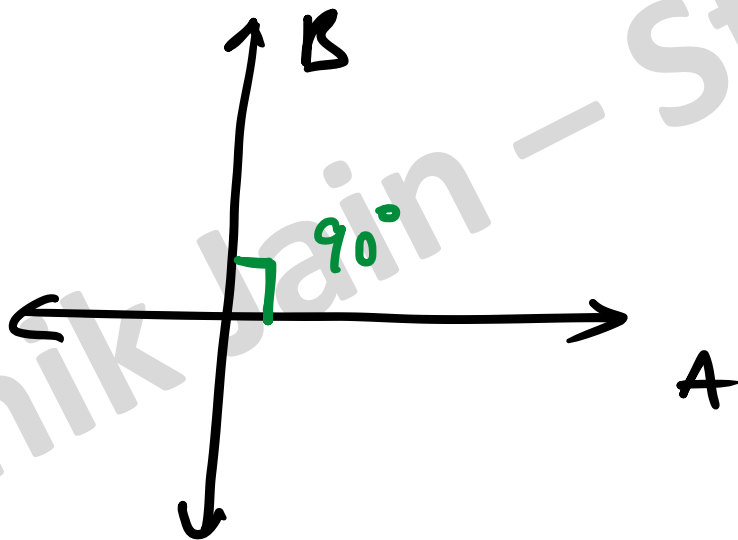
Ray

⑤



Parallel

⑥

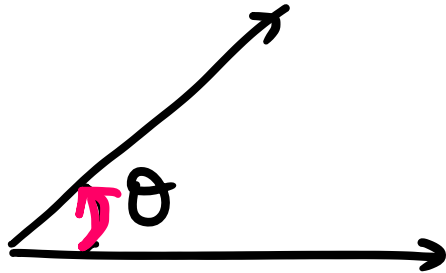


A is perpendicular to B

$\Rightarrow A \perp B$

#

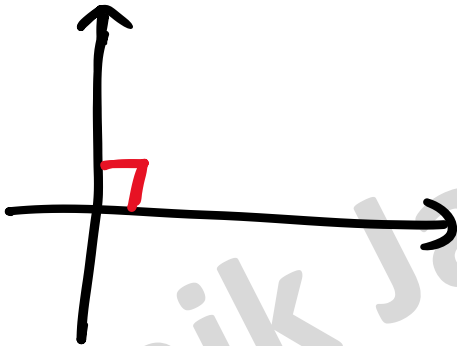
θ



$\Rightarrow$

$$0 < \theta < 90$$

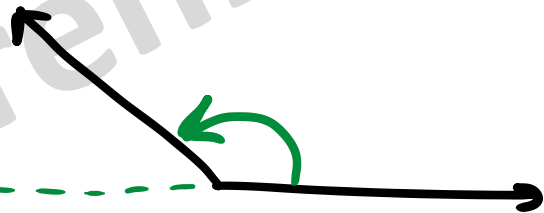
$\Rightarrow$ Acute angle



$\Rightarrow$

$$\theta = 90^\circ$$

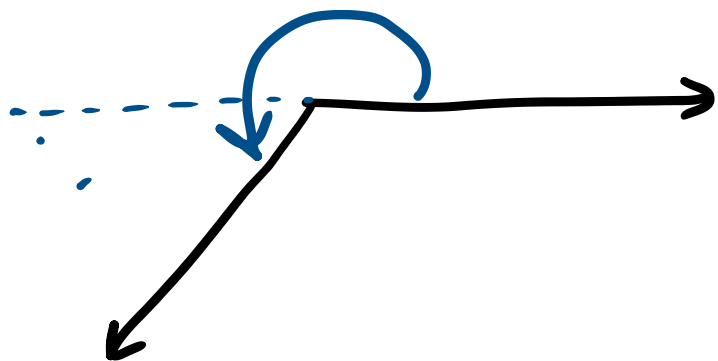
$\Rightarrow$ Right angle



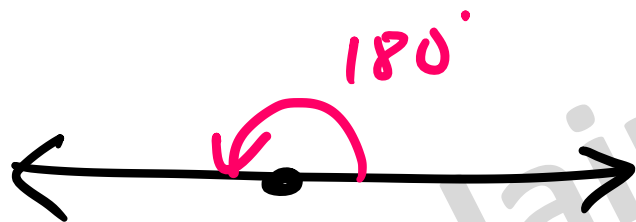
$\Rightarrow$

$$90 < \theta < 180$$

$\Rightarrow$ obtuse angle

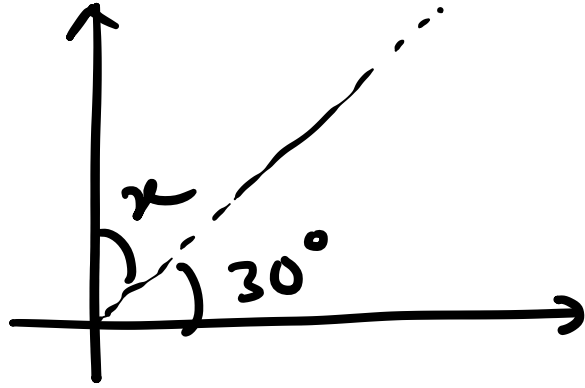


$\Rightarrow 180 < \theta < 360 \Rightarrow \text{Reflex angle}$



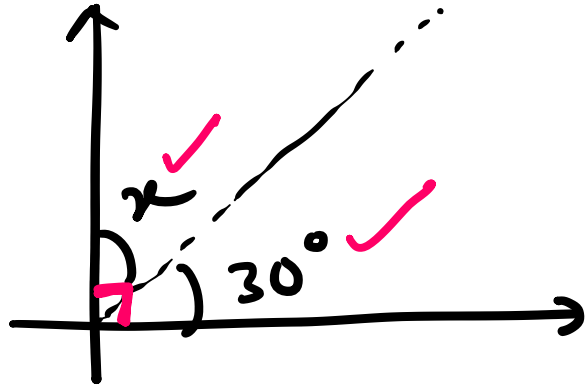
$\Rightarrow \theta = 180^\circ \Rightarrow \text{straight line}$

Q]



$x =$ cannot be determined

Q]



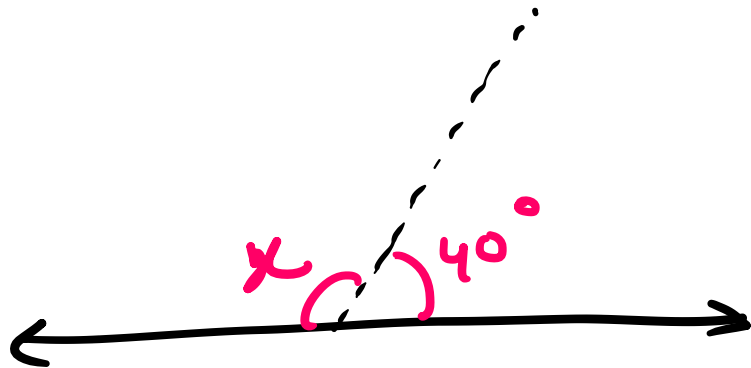
[Complementary angles]

~~$x =$ cannot be determined~~

$$90^\circ = x + 30$$

$$x = 90 - 30 = 60^\circ //$$

9]

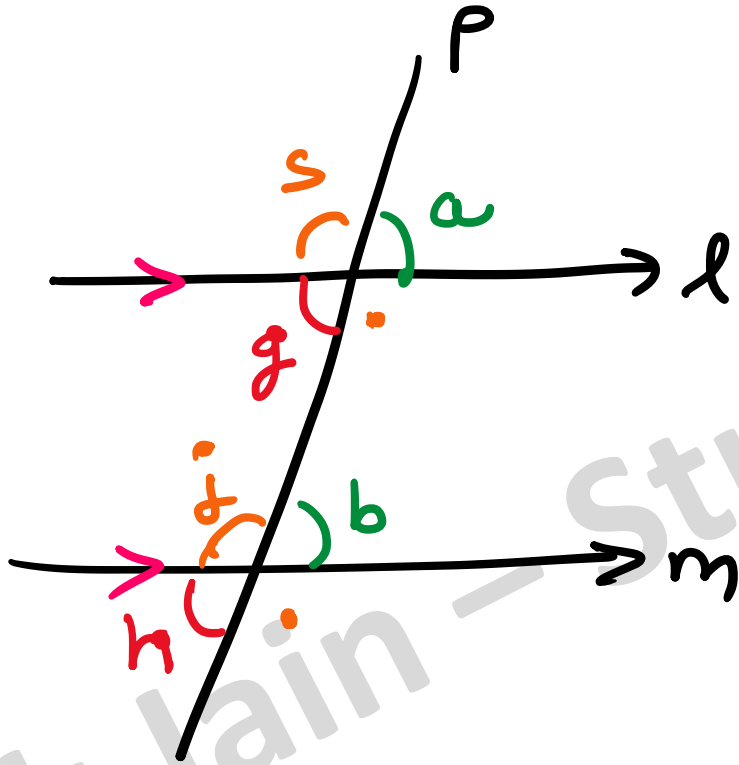


Supplementary
angles

$$\Rightarrow x + 40 = 180$$

$$\therefore x = 180 - 40 = 140^\circ$$

#



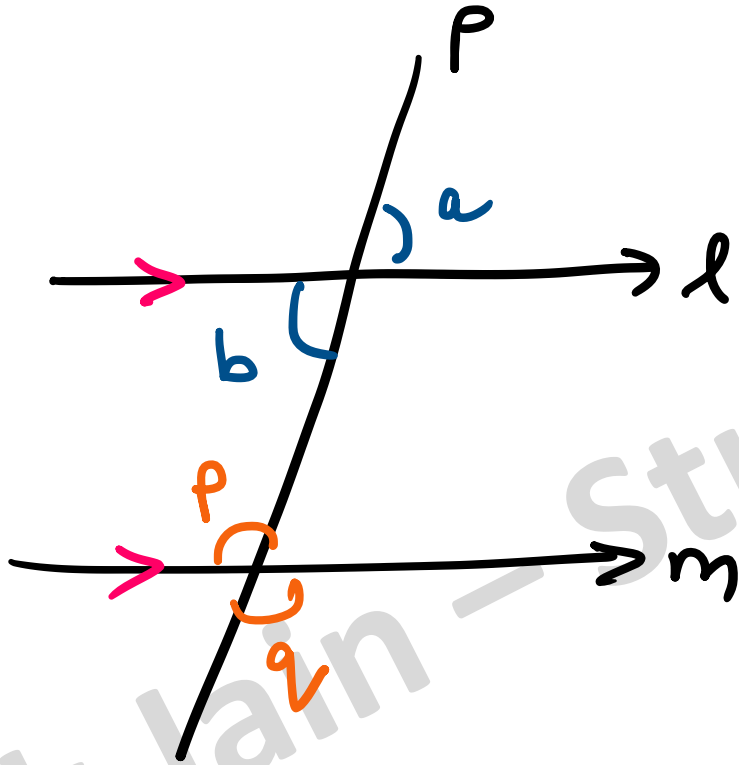
corresponding
angle

$$\angle a = \angle b$$

$$\angle g = \angle h$$

$$\angle s = \angle j$$

#

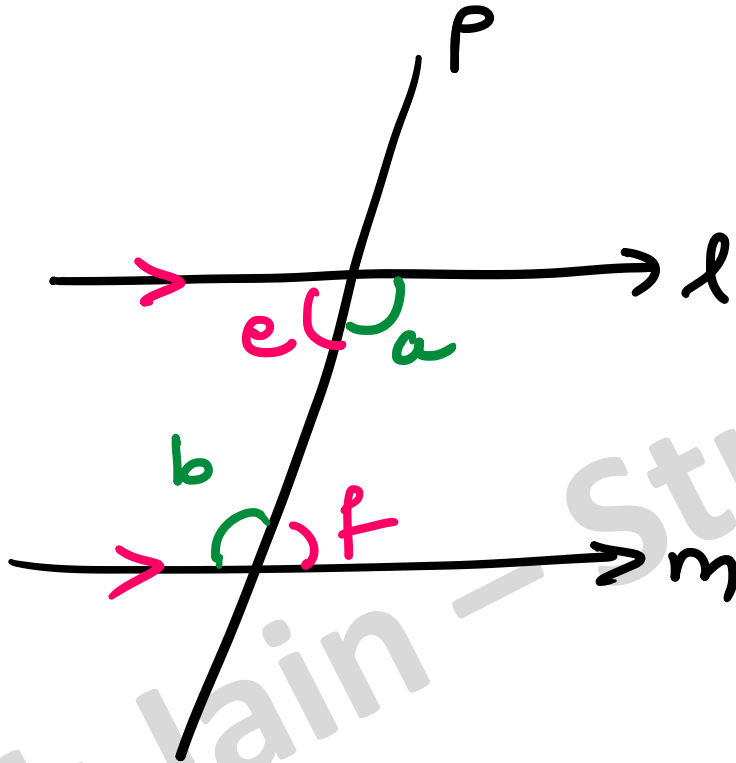


Vertically
opposite angles

$$\angle a = \angle b$$

$$\angle p = \angle q$$

#

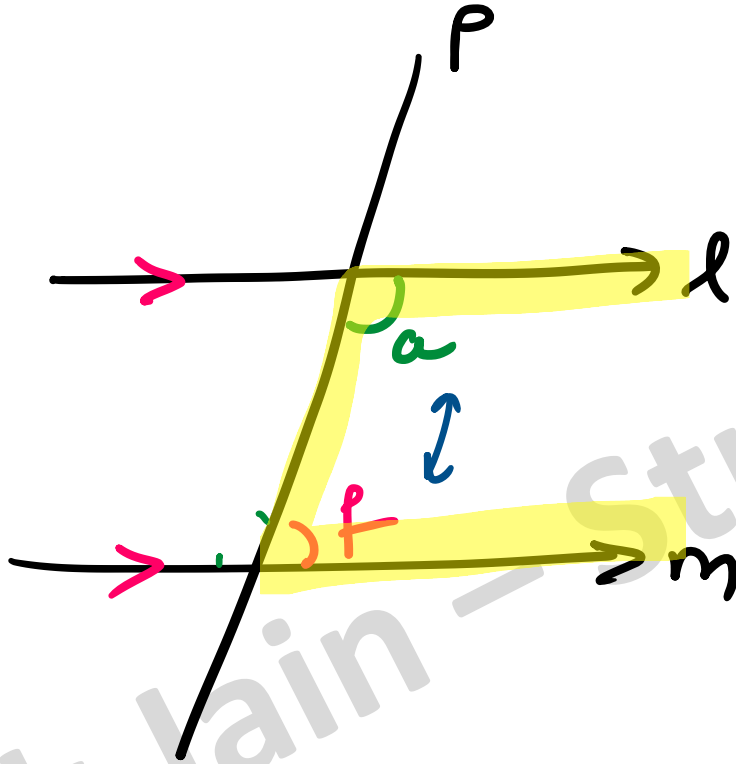


Alternate Angles

$$\angle a = \angle b$$

$$\angle e = \angle f$$

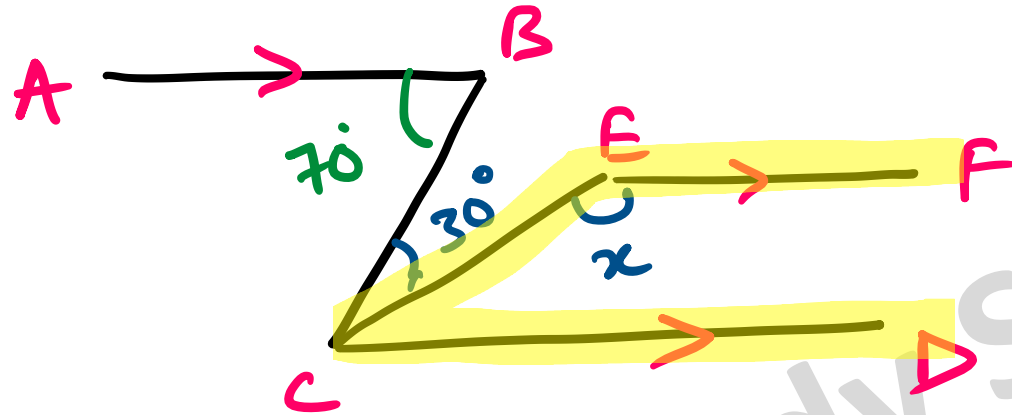
#



$$\angle a + \angle f = 180^\circ$$

Shrenik Jain - Study Simplified

8]



solⁿ:

① $\angle ABC = \angle BCD = \underline{70^\circ}$

② $\angle BCD = \angle BCE + \angle ECD$

$$70^\circ = 30^\circ + \angle ECD$$

$$\angle ECD = 40^\circ$$

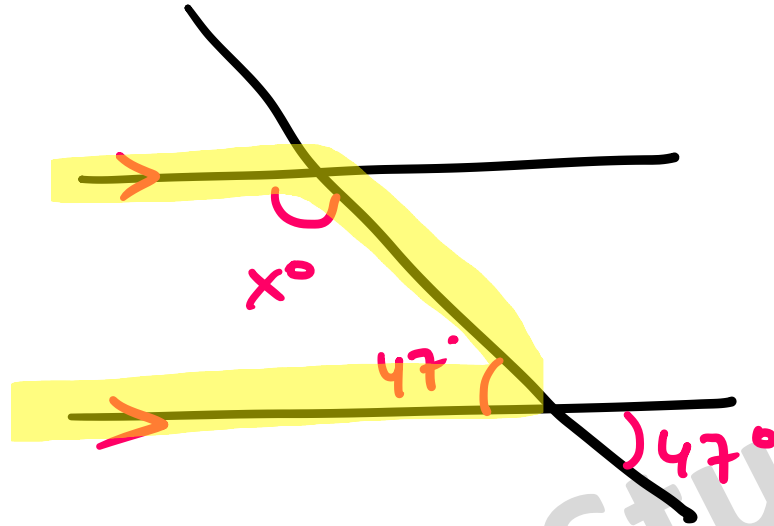
③

$$x + \angle ECD = 180^\circ$$

$$x + 40 = 180$$

$$\boxed{x = 140^\circ}$$

Q]



Solⁿ :

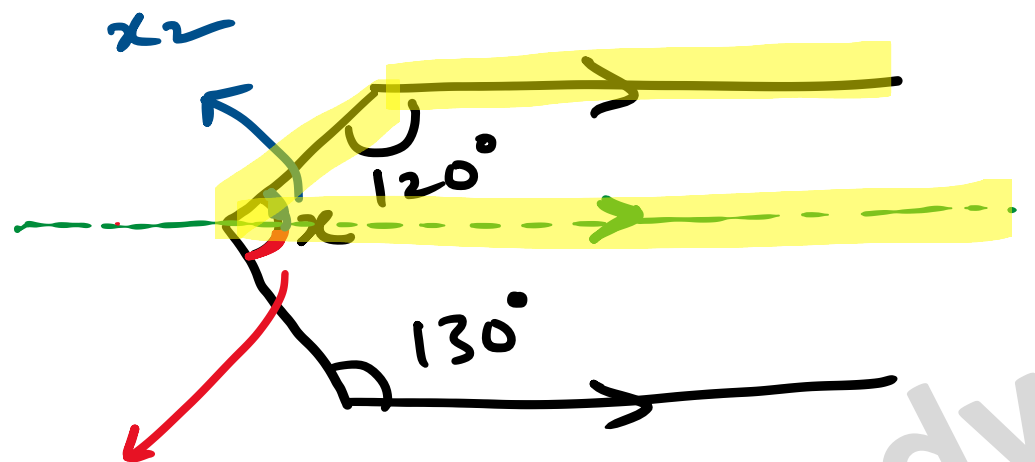
$$\angle x + \angle 47 = 180^\circ$$

$$x = 180 - 47$$

$$= 133^\circ //$$

Q]

(M1)



$$\begin{aligned}x &= x_1 + x_2 \\&= 50 + 60 \\&= 110^\circ\end{aligned}$$

$$120 + x_2 = 180$$

$$x_2 = 180 - 120$$

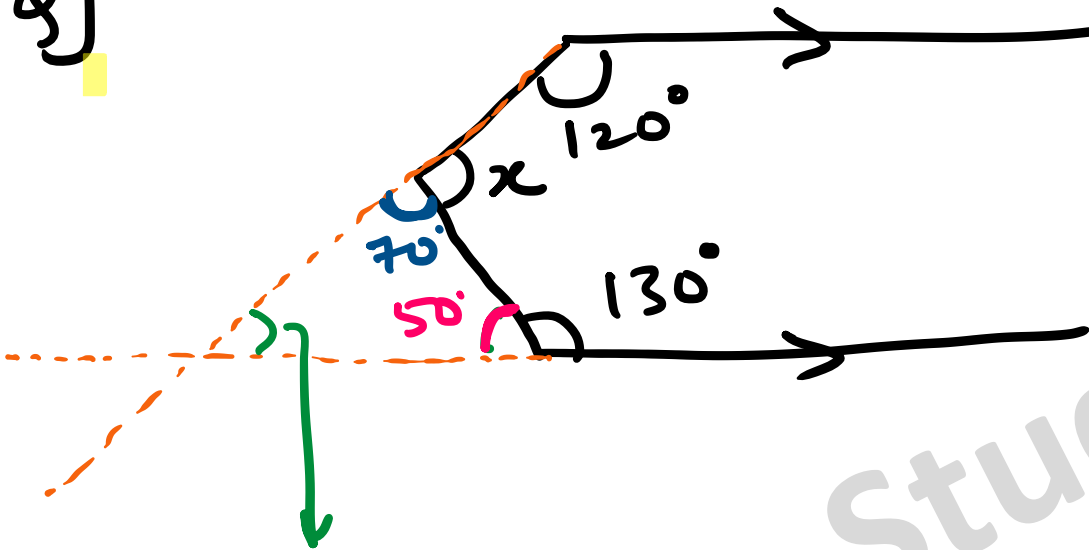
$$x_2 = 60^\circ$$

$$x_1 + 130^\circ = 180$$

$$x_1 = 180 - 130$$

$$x_1 = 50$$

Q]



$$\theta + 120 = 180$$

$$\theta = 60$$

$$70 + x = 180$$

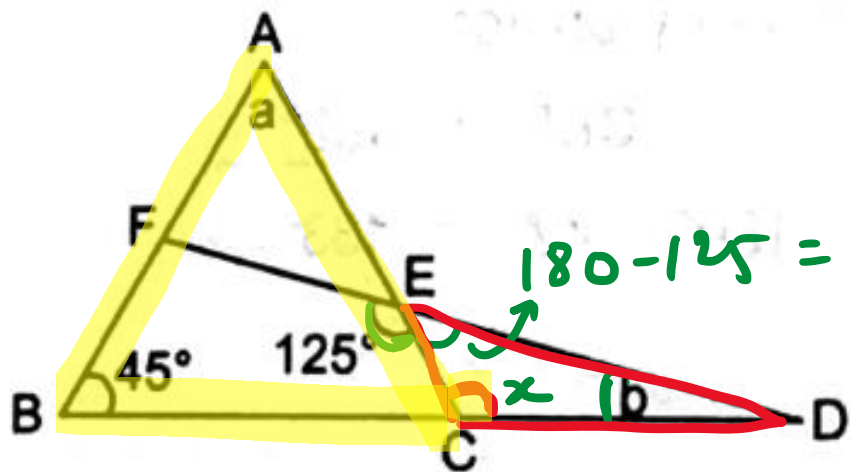
$$x = 110$$

$$? + 60 + 50 = 180$$

$$\begin{aligned} ? &= 180 - 60 - 50 \\ &= 70 \end{aligned}$$



M2



$$a + b = ?$$

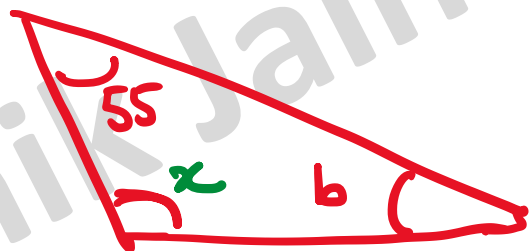
$$180 - 125 = 55^\circ$$

$$x = \underline{45 + a}$$

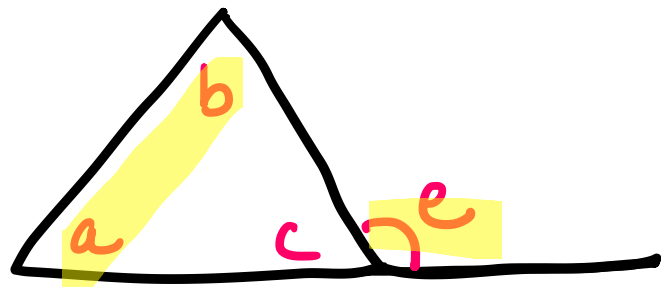
$$55 + \underline{x} + b = 180^\circ$$

$$\Rightarrow 55 + (45 + a) + b = 180$$

$$a + b = 80^\circ //$$



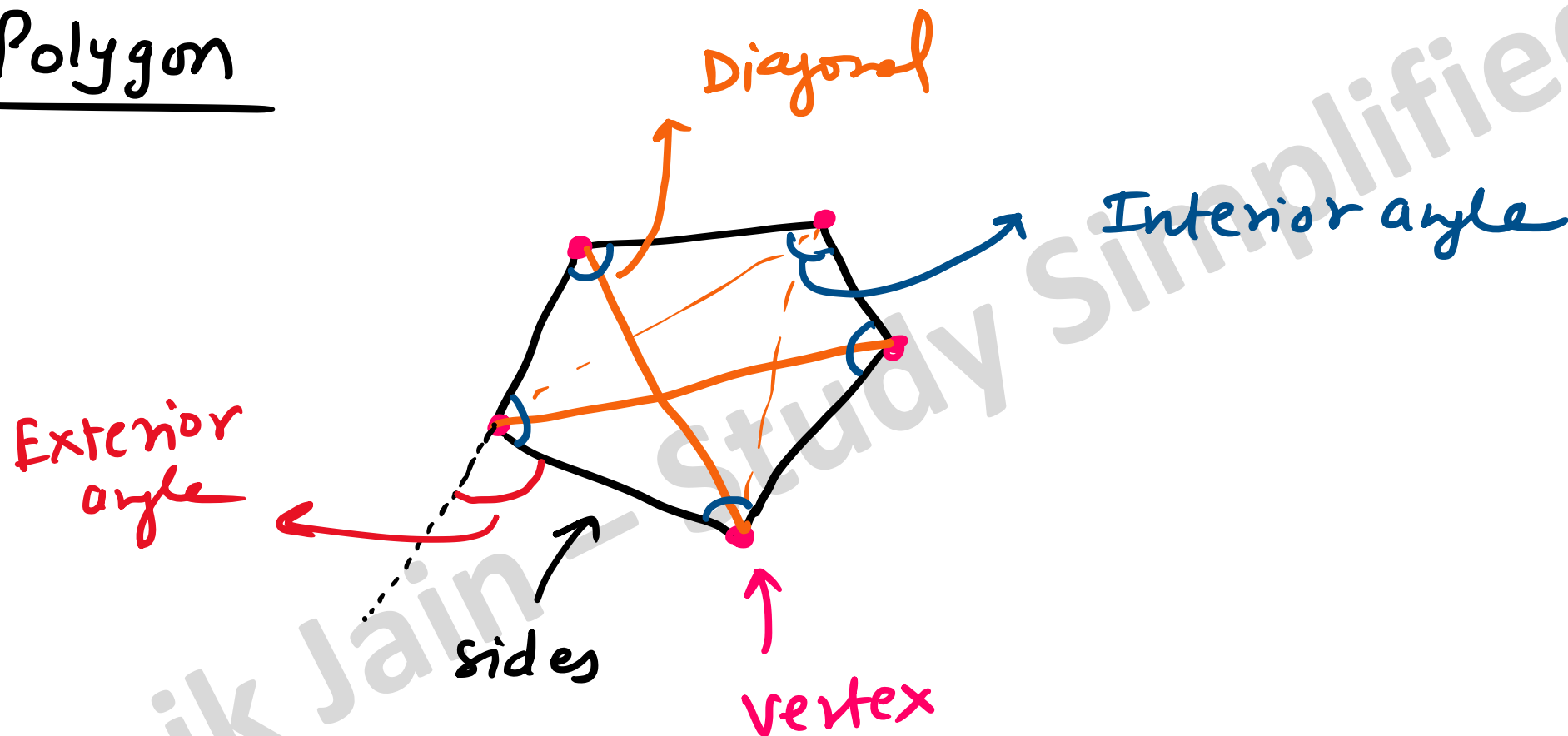
#



$$\angle e = \angle a + \angle b$$

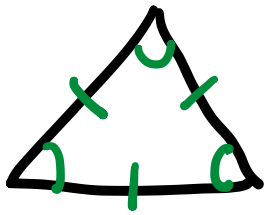
Shrenik Jain — Study Simplified

Polygon

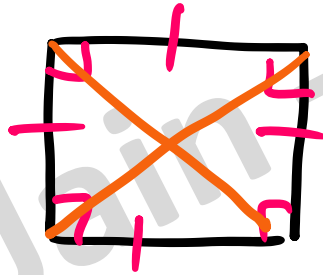


Regular polygon

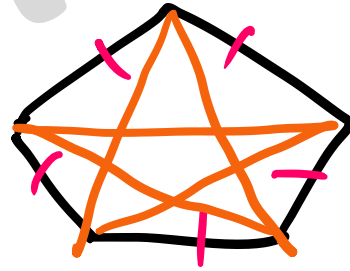
- Equal sides
- Equal interior angles



(Triangle)



(Square)



(Pentagon)

Hexa = 6

Hepta = 7

Octa = 8

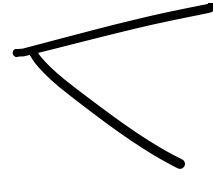
n-sided regular polygon

- ✓ ① Sides = n
- ✓ ② Diagonals = $\frac{n}{2}(n-3)$
- ✓ ③ Σ of interior angle of polygon = $(n-2)180^\circ$
- ④ Σ of exterior angle of polygon = 360°

Each interior angle $\rightarrow \left(180 - \frac{360}{n}\right)$

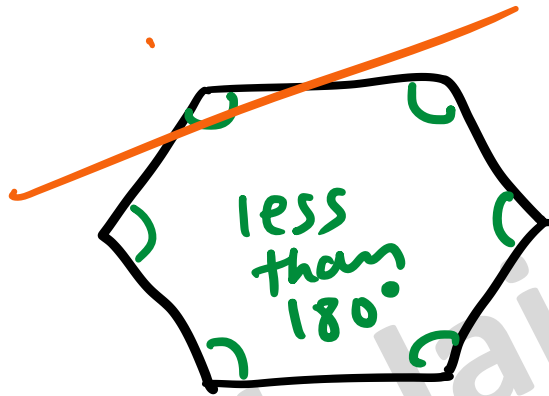
Each exterior angle $\rightarrow \frac{360}{n}$

Polygon

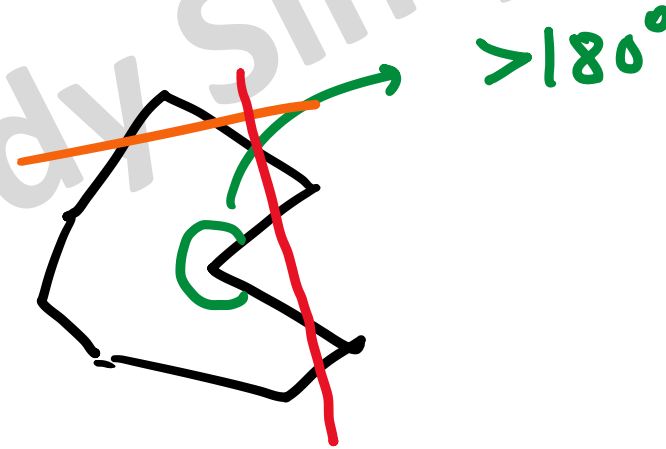


Convex

Concave



Convex



Concave

• . 9] How many diagonals does a 12 sided polygon have?

Solⁿ:

$$n = 12$$

$$\text{Diagonals} = \frac{n}{2} (n-3)$$

$$= \frac{12}{2} (12-3)$$

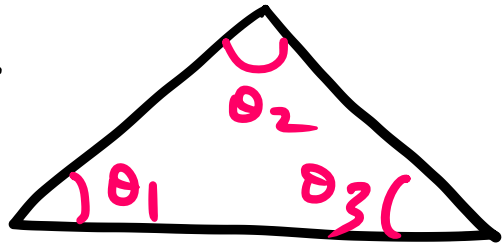
$$= 54 //$$

Q] Sum of all interior angle of regular polygon is 3240. What is each exterior angle?

Solⁿ : (1) $(n-2) 180 = 3240$

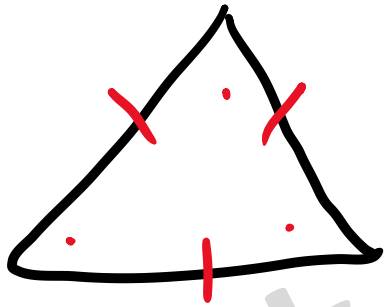
$$n = 20$$

(2) $\frac{360}{n} = \frac{360}{20} = 18^\circ //$

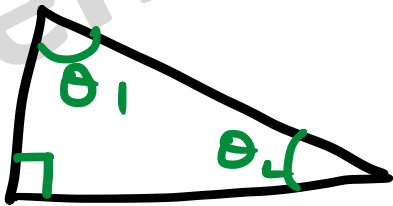


Acute angle Δ

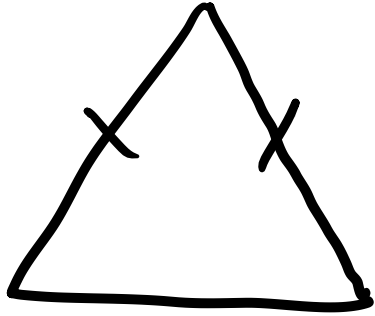
$$[\theta_1, \theta_2, \theta_3] < 90^\circ$$



Equilateral Δ



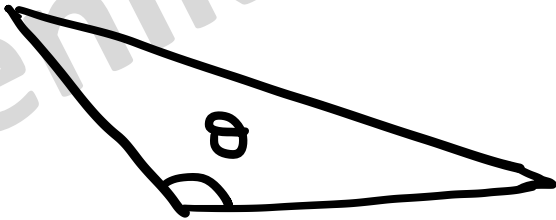
Right angle Δ



Isosceles $\triangle$



Scalene $\triangle$



obtuse angled $\triangle$

A wall, rectangular in shape, has a perimeter of 72 m. If the length of its diagonal is 18m, what is the area of the wall?

(a) 224m^2

(b) 486m^2

(c) 572m^2

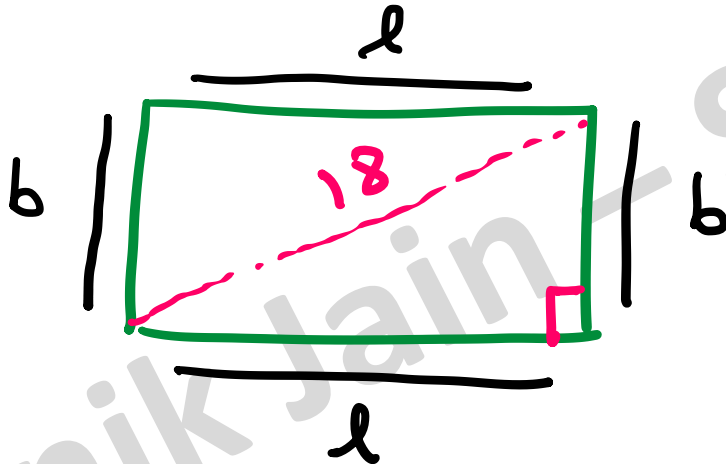
(d) 606m^2

[ESE-2018 (Common Paper)]

(3)

$$l^2 + b^2 = 18^2$$

(1)



(2) Perimeter = $l + l + b + b = 72$

$$\Rightarrow 2(l + b) = 72 \Rightarrow l + b = 36$$

$$\Rightarrow \boxed{l+b=36}$$

$$(l+b)^2 = 36^2$$

$$\cancel{l^2} + 2lb + \cancel{b^2} = 1296$$

$$324 + 2lb = 1296$$

$$A = \boxed{lb = 486}$$

A rectangle becomes a square when its length and breadth are reduced by 10 m and 5 m respectively. During this process, the rectangle loses 650 m² of area. What is the area of the original rectangle in square meters?

~~(a) 1125~~

✓ (b) 2250

~~(c) 2954~~

(d) 4500

[GATE-2018 (ME Morning Session)]

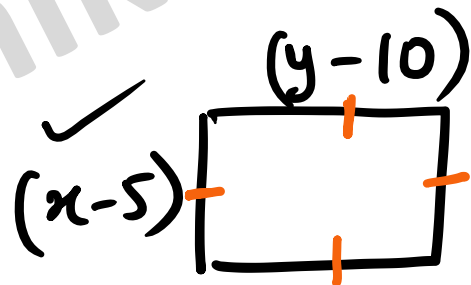
$$\rightarrow xy = 45 \times 50 \\ = 2250 //$$

①



$$\Rightarrow A(\square) = xy$$

②



$$\Rightarrow A(\square) = (x-5)^2 \\ (y-10) = (x-5)$$

$$A(\square) - A(\square) = 650$$

$$xy - (x-5)^2 = 650$$

$$x(5+x) - (x^2 - 10x + 25) = 650$$

$$\Rightarrow 15x - 25 = 650$$

$$\Rightarrow \boxed{x = 45}$$

$$y = 5 + x$$

$$\boxed{y = 50}$$

✓ A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is _____.

(a) 1.43

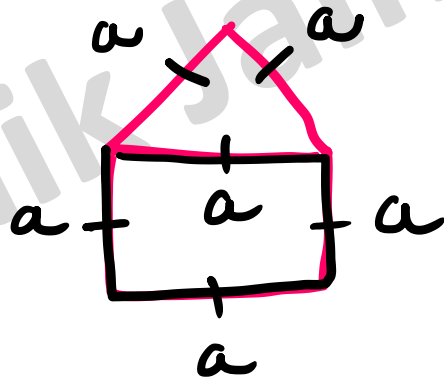
(b) 2.06

(c) 2.68

(d) 2.88

[GATE-2016]

①

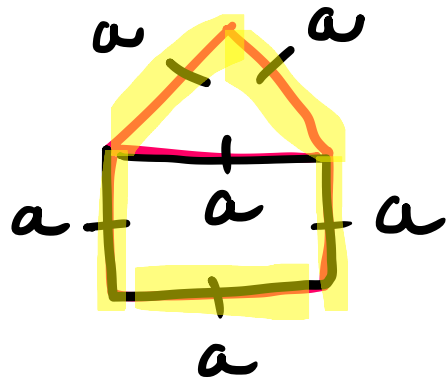


$$6a = 6$$

$$a = 1$$

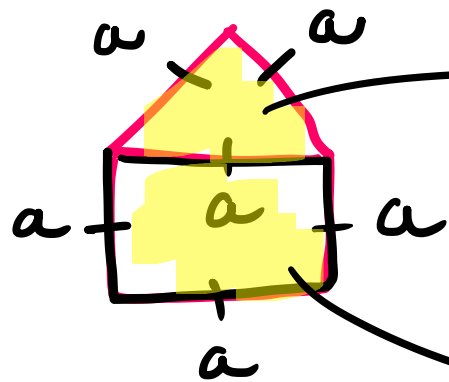
XXX

[wrong]



$$\begin{aligned}\Rightarrow \text{Perimeter} &= \Sigma \text{ of all sides} \\ &= a + a + a + a + a \\ &= 5a = 6\end{aligned}$$

$$\Rightarrow \boxed{a = 1.2 \text{ m}}$$



$$A(\Delta) = \frac{\sqrt{3}}{4} a^2$$

+

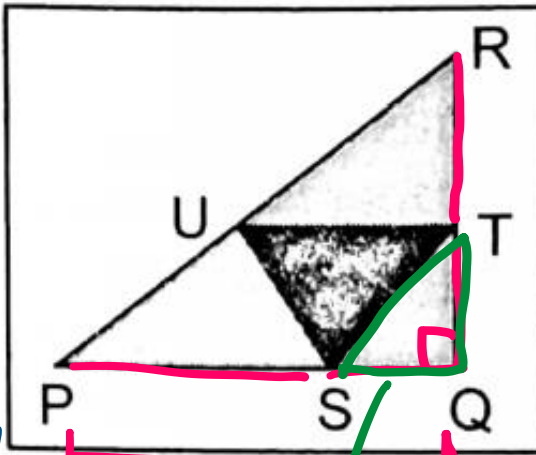
$$A(\square) = a^2$$

Area
of
window

$$= \frac{\sqrt{3}}{4} a^2 + a^2$$

$$= 2.063 \text{ m}^2 //$$

In the given figure angle Q is a right angle, PS : QS
= 3:1, RT:QT = 5:2 and PU:UR = 1:1. If area of
triangle QTS is 20 cm^2 , then the area of triangle
PQR in cm^2 is _____.



[GATE-2015, ME, EC]

$$\frac{PS}{QS} = \frac{3}{1}$$

$$PS = 3QS$$

$$RT = \frac{5}{2} QT$$

$$A = \frac{1}{2} (SQ) (QT) = 20$$

$$(SQ) (QT) = 40$$

$$\begin{aligned} A &= \frac{1}{2} \times PQ \times RQ \\ &= \frac{1}{2} (\cancel{PS} + \overset{3QS}{SQ}) (\cancel{RT} + \overset{2.5QT}{TQ}) \\ &= \frac{1}{2} (4QS) (3.5QT) \\ &= 7 (QS) (QT) \\ &= 7 \times 40 \\ &= 280 // \end{aligned}$$